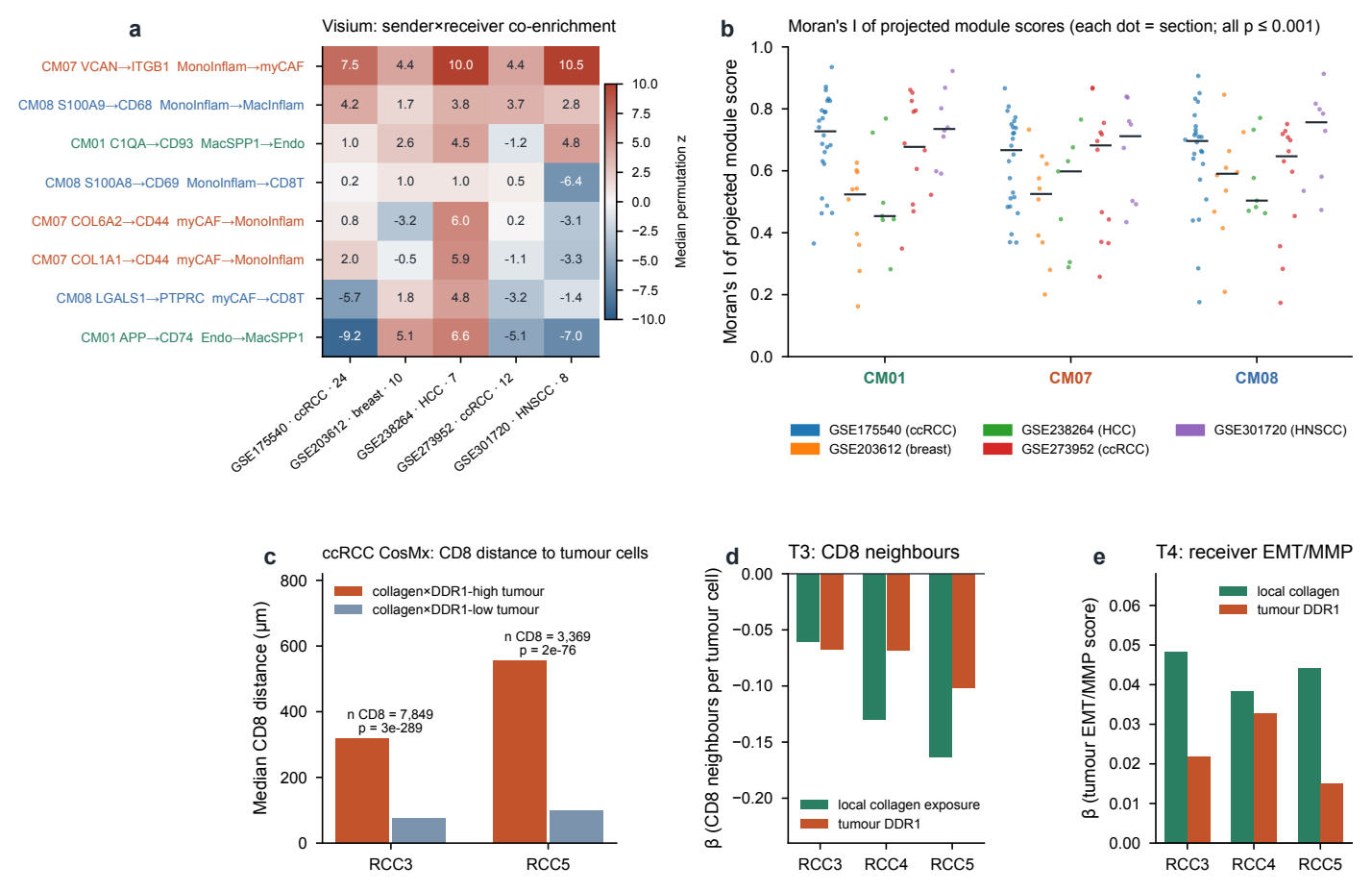

a
Visium: sender×receiver co-enrichment
CM07 VCAN→ITGB1 MonoInflam→myCAF
CM08 S100A9→CD68 MonoInflam→MacInflam
CM01 C1QA→CD93 MacSPP1→Endo
CM08 S100A8→CD69 MonoInflam→CD8T
CM07 COL6A2→CD44 myCAF→MonoInflam
CM07 COL1A1→CD44 myCAF→MonoInflam
CM08 LGALS1→PTPRC myCAF→CD8T
CM01 APP→CD74 Endo→MacSPP1
7.5 4.4 10.0 4.4 10.5
4.2 1.7 3.8 3.7 2.8
1.0 2.6 4.5 -1.2 4.8
0.2 1.0 1.0 0.5 -6.4
0.8 -3.2 6.0 0.2 -3.1
2.0 -0.5 5.9 -1.1 -3.3
-5.7 1.8 4.8 -3.2 -1.4
-9.2 5.1 6.6 -5.1 -7.0
Median permutation z
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
GSE175540 · ccRCC · 24
GSE203612 · breast · 10
GSE238264 · HCC · 7
GSE273952 · ccRCC · 12
GSE301720 · HNSCC · 8
b
Moran's I of projected module scores (each dot = section; all p ≤ 0.001)
Moran's I of projected module score
1.0
0.8
0.6
0.4
0.2
0.0
CM01
CM07
CM08
GSE175540 (ccRCC)
GSE203612 (breast)
GSE238264 (HCC)
GSE273952 (ccRCC)
GSE301720 (HNSCC)
c
ccRCC CosMx: CD8 distance to tumour cells
collagen×DDR1-high tumour
collagen×DDR1-low tumour
Median CD8 distance (µm)
800
600
400
200
0
n CD8 = 7,849
p = 3e-289
n CD8 = 3,369
p = 2e-76
RCC3
RCC5
d
T3: CD8 neighbours
β (CD8 neighbours per tumour cell)
0.00
−0.05
−0.10
−0.15
−0.20
local collagen exposure
tumour DDR1
RCC3
RCC4
RCC5
e
T4: receiver EMT/MMP
local collagen
tumour DDR1
β (tumour EMT/MMP score)
0.06
0.05
0.04
0.03
0.02
0.01
0.00
RCC3
RCC4
RCC5